

Azure Key Vault

Distributed Application Development
in the Cloud – NET 031

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Agenda

- What is Azure Key Vault.
- Demo
 - Setting up the Key Vault.
 - Creating a Secret.
 - Accessing keys from a .Net API.
 - Setting up Config variables after deploying into azure app service.

What is Azure Key Vault

- In enterprise grade applications it is paramount important to safeguard different keys, secrets, certificates etc.
- Loosing these will make your app vulnerable and may cause drastic negative effects.
- Think about how you store your connection strings in your local environment. Does it look okay to you.
- Storing these credentials in plain text would be a huge mistake. You might even commit it to your source control accidentally.
- There are different approaches to safeguard your secrets and it really depends on your scenario.
- Azure Key Vault is a cloud-based solution provided by Azure which even supports HSM (Hardware Security Modules).
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What is Azure Key Vault Contd.

- Azure Key Vault helps to safeguard cryptographic keys and other secrets used by cloud apps and services. (We can retrieve a key even from a console application)
- Azure Key Vault:
 - Increase security and control over keys and passwords.
 - Applications have no direct access to keys.
 - Use FIPS 140-2 Level 2 and Level 3 validated HSMs.
 - Create and import encryption keys in minutes.And many more features.....



Creating an Azure Key Vault

- Creating an Azure Key Vault is easy, and you can use your free account.

Home > Key vaults >

Create a key vault

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

[Create new](#)

Instance details

Key vault name * ⓘ

Region *

Pricing tier * ⓘ



Breakout Rooms

Hands-On/ Lab Exercise

Every student is expected to complete this step before moving forward.

Exploring the Key Vault

 **dotnet031vault** | Secrets ...
Key vault

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Access policies
Events

Objects

- Keys
- Secrets**
- Certificates

« + Generate/Import ↻ Refresh ⬆ Restore Backup

Name	Type
QueueStorageSettings--Connection...	
QueueStorageSettings--ConStr	
QueueStorageSettings--OrderQueu...	



Breakout Rooms

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Demo



Breakout Rooms

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THANK YOU.



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